

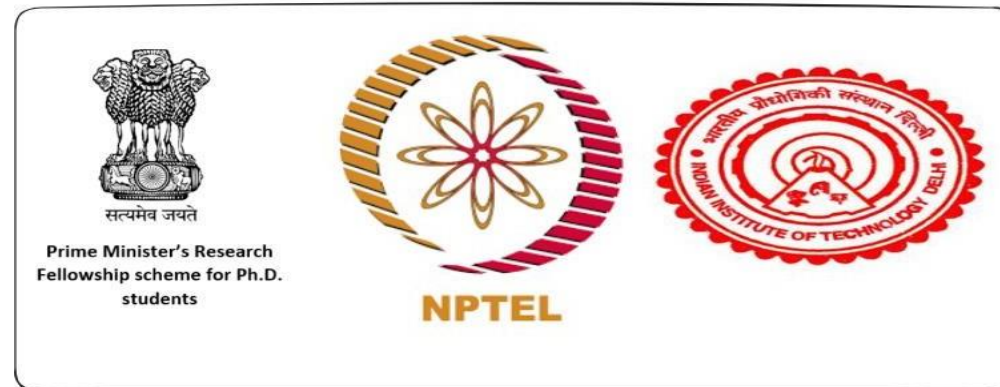
Interaction Session 9

on

Cloud Computing
noc25-cs11

by

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3rd March 2025

Indian Institute of Technology, Delhi

Points to Remember:

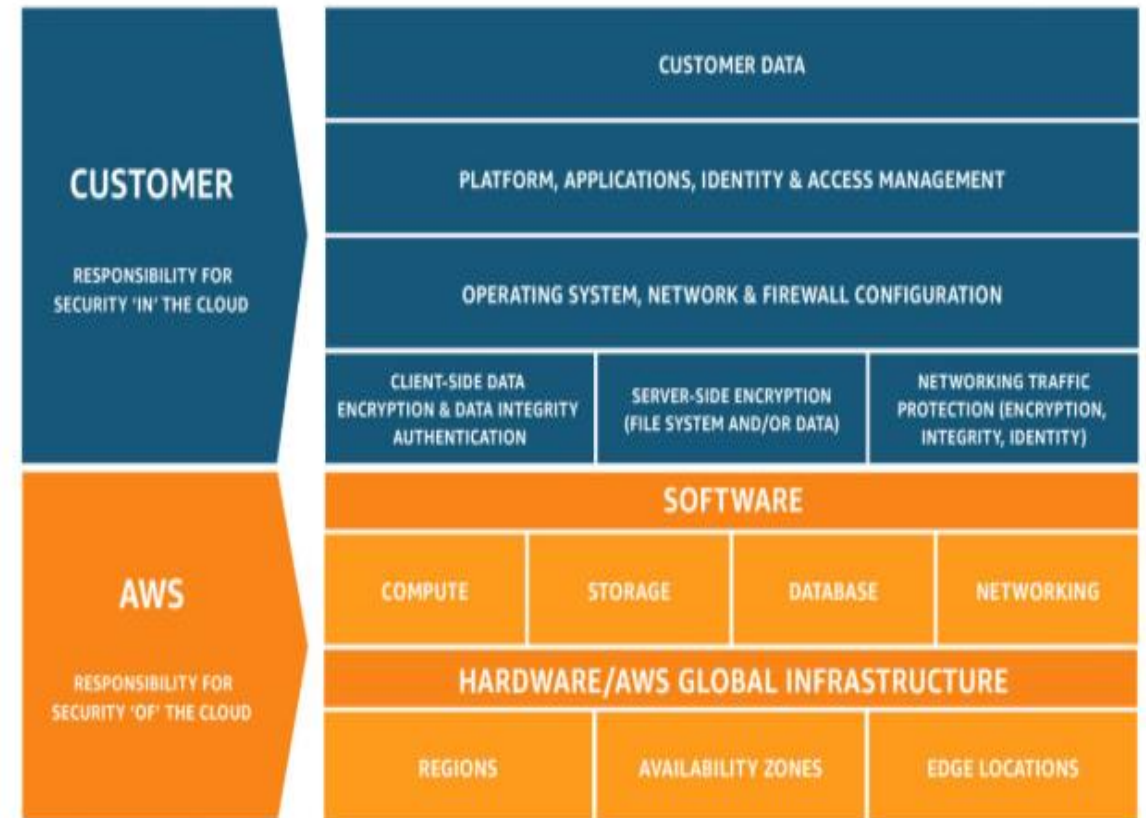
- Total 6 more Discussion sessions to be taught lectures and Tutorials
- Any specific problem, please upload in the discussion forum, I will try to reply within 24 hours
- PPT and Video will be shared with everyone
- Assignments will not be solved but similar problems will be solved.

TOPICS TO BE COVERED

1. Cloud Computing Security
2. Security Groups in AWS
3. Hands-on S3

Security in Cloud Computing

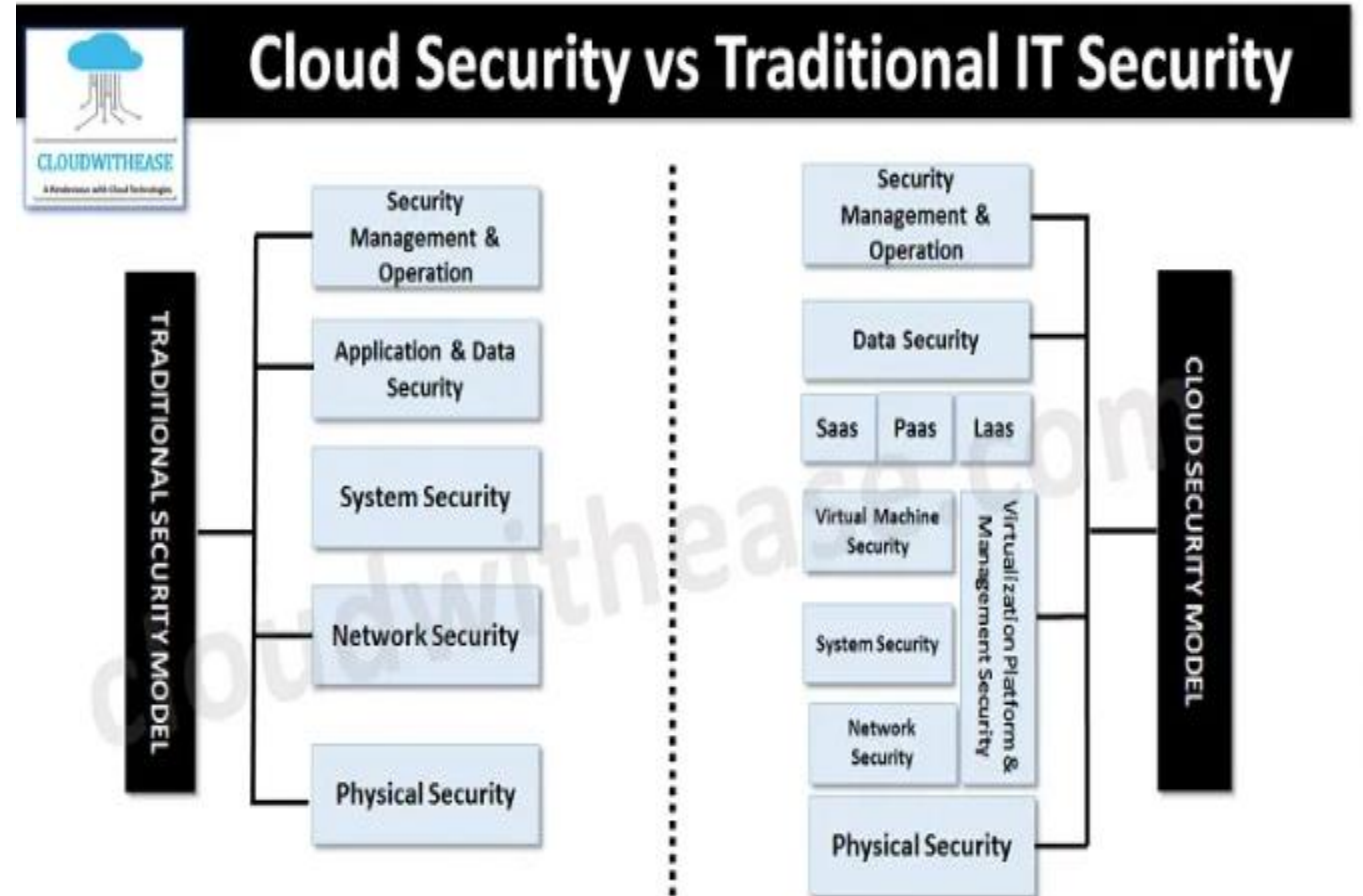
- Cloud computing involves outsourcing data and computation, providing benefits such as:
 - Scalability: Resources can expand or contract based on demand.
 - On-Demand Provisioning: Resources are available when needed without upfront investments.
 - Cost Savings: Pay-as-you-go pricing eliminates the need for expensive on-premises infrastructure².
 - Security in Cloud Computing
- Cloud security focuses on the following principles:
 - Confidentiality: Ensures that sensitive data is accessible only to authorized users through encryption, access controls, and authentication mechanisms.
 - Integrity: Protects data from unauthorized modification or tampering using tools like hashing, digital signatures, and audits.
 - Availability: Ensures reliable access to data and systems through redundancy, fault tolerance, and disaster recovery mechanisms.
 - Authentication: Verifies the identity of users accessing the system.
 - Non-Repudiation: Ensures that actions cannot be denied by their originators using cryptographic methods like digital certificates



LINK : <https://aws.amazon.com/compliance/shared-responsibility-model/>

What Makes Cloud Security Different from Traditional IT Security?

- Shared Responsibility Model: In cloud environments, security responsibilities are divided between the cloud provider (infrastructure-level security) and the customer (data and application-level security)
- Scalability: Cloud security must handle dynamic resource allocation, unlike traditional setups limited by physical infrastructure
- Multi-Tenancy Risks: Multiple customers share the same infrastructure in the cloud, requiring strict isolation to prevent data breaches
- Encryption in Transit and at Rest: Cloud environments emphasize securing data during transmission over the internet and while stored in remote locations
- Centralized Management: Cloud security provides a unified dashboard for managing multi-cloud environments, unlike fragmented traditional systems



How Do Confidentiality, Integrity, and Availability Apply to Cloud Services?

- **Confidentiality**

Protects sensitive data from unauthorized access using encryption (e.g., AES), multi-factor authentication (MFA), and access controls.

Example: A healthcare provider encrypts patient records to comply with privacy regulations like HIPAA⁵⁸.

- **Integrity**

Ensures data accuracy and trustworthiness by preventing unauthorized modifications.

Tools include digital signatures, hashing algorithms, and regular audits.

Example: Banks use integrity measures to ensure financial transactions are not tampered with during processing⁵¹³.

- **Availability**

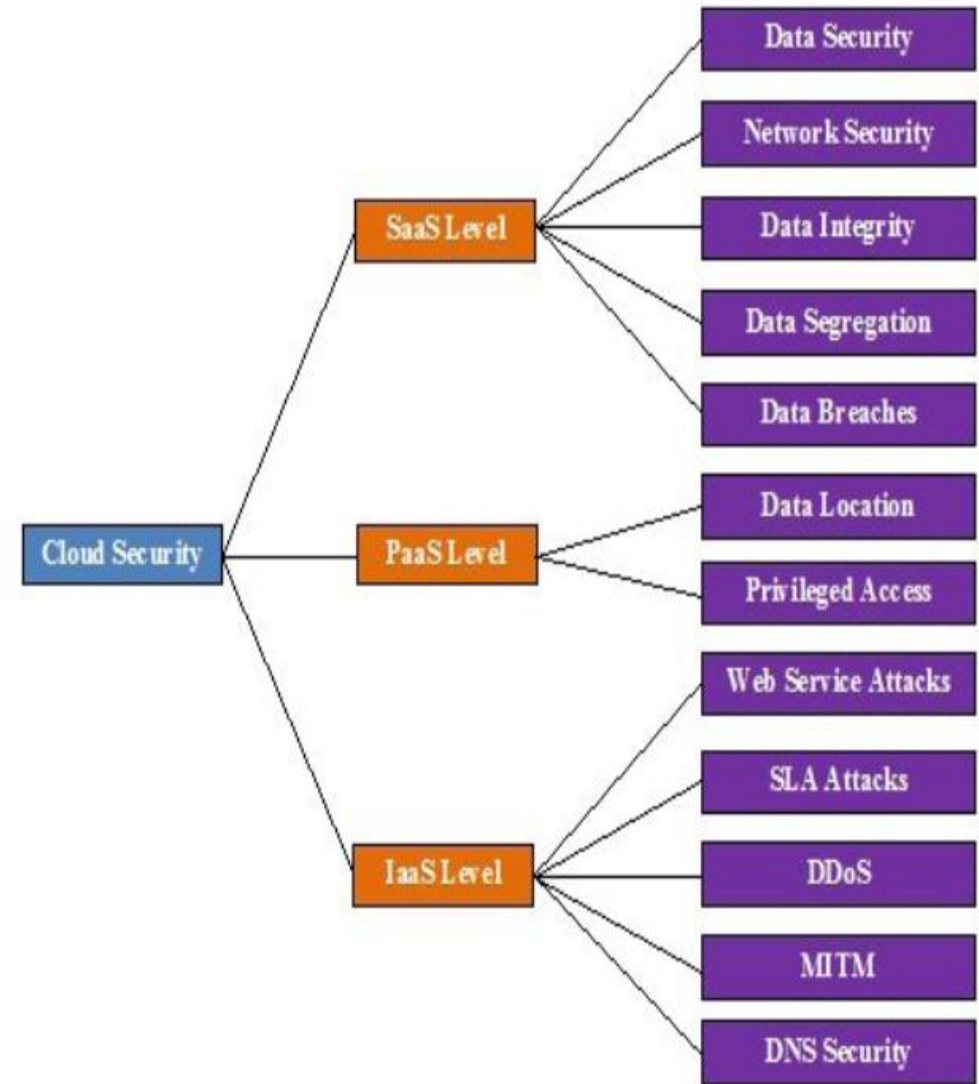
Ensures uninterrupted access to services through redundancy (e.g., backup servers) and disaster recovery plans.

Example: E-commerce platforms rely on load balancing to maintain availability during high traffic periods like Black Friday sales⁵⁹.

Key Security Concepts and Threats

Types of Attacks

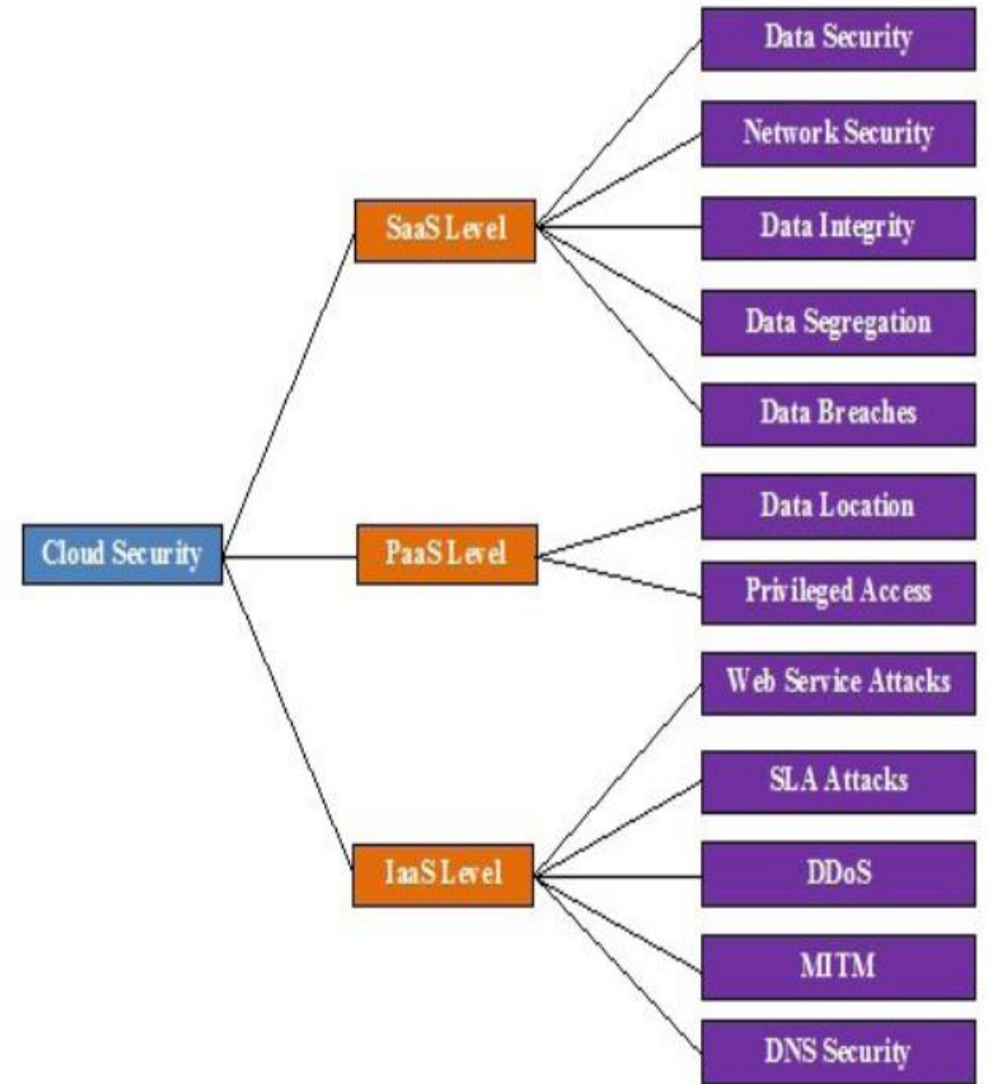
- Interruption:
 - Targets availability by disrupting services or systems, making them temporarily or permanently unavailable.
 - Examples: Denial-of-Service (DoS) attacks, hardware destruction, or erasing critical files.
 - Interception:
 - Targets confidentiality by unauthorized access to data during transmission or storage.
 - Examples: Packet sniffing, man-in-the-middle (MitM) attacks, or wiretapping.
 - Modification:
 - Targets integrity by altering data or systems without authorization.
 - Examples: Data tampering, code injection, or unauthorized system changes.
 - Fabrication:
 - Targets authenticity by creating false data or transactions to mislead systems or users.
 - Examples: Spoofing, fake user accounts, or counterfeit transactions
- ### Passive vs Active Attacks
- Passive Attacks:
 - Focus on covertly gathering information without altering the system.
 - Examples: Eavesdropping on network traffic, traffic analysis, or monitoring unencrypted communications.
 - Characteristics: Harder to detect because they do not leave noticeable traces.
 - Active Attacks:
 - Involve direct interaction with the system to disrupt, alter, or manipulate data.
 - Examples: Masquerade (impersonating a legitimate user), replay attacks (reusing intercepted data), and denial-of-service (DoS) attacks.
 - Characteristics: Easier to detect due to noticeable disruptions in system functionality.



Key Security Concepts and Threats

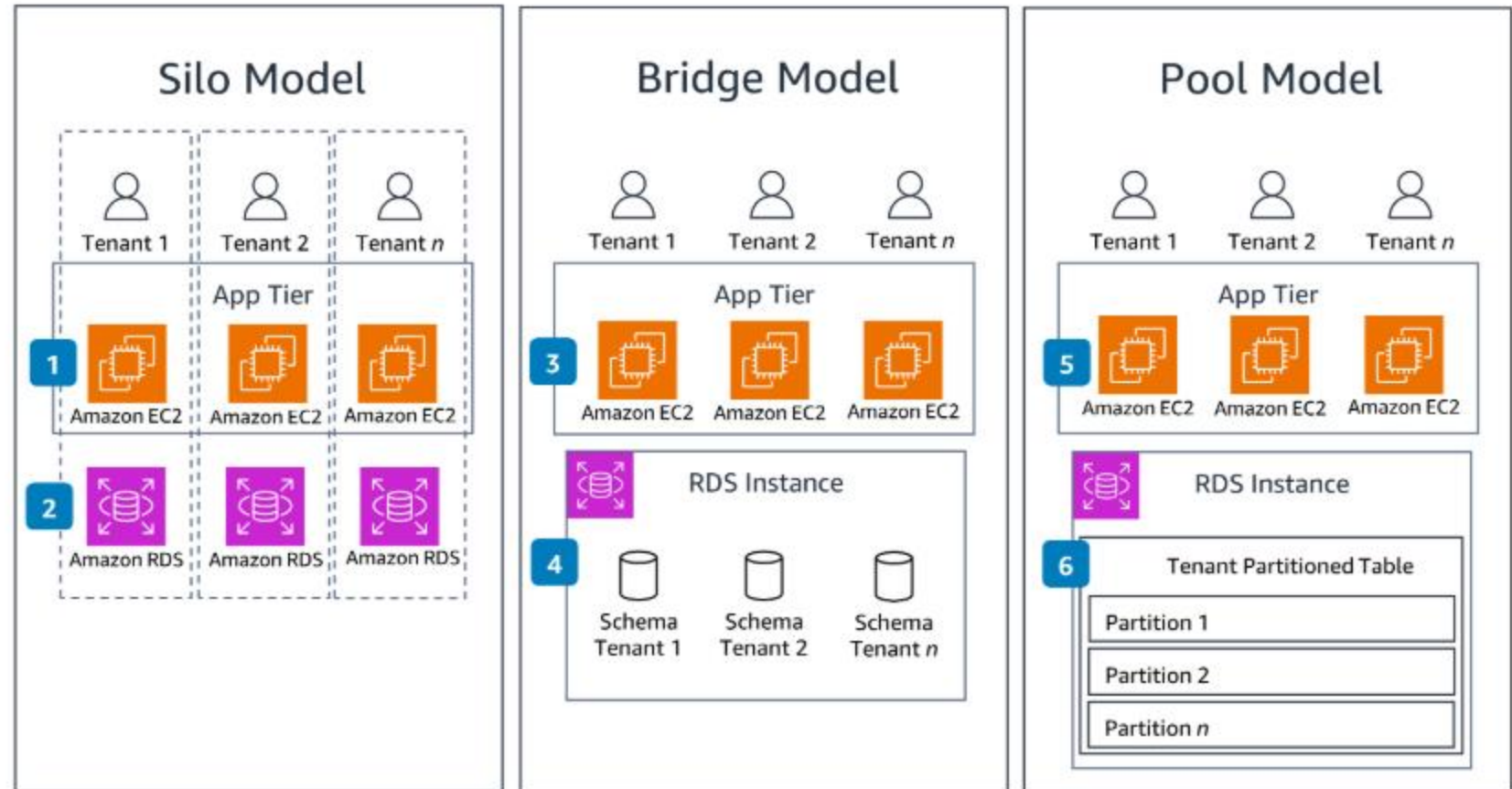
Classes of Threats

- Disclosure (Snooping):
 - Unauthorized access to confidential data.
 - Example: Intercepting sensitive emails.
- Disruption (Modification):
 - Altering data or systems to cause errors or failures.
 - Example: Changing database records to manipulate outcomes.
- Usurpation (Spoofing):
 - Gaining unauthorized control by impersonating legitimate entities.
 - Example: Phishing attacks where attackers pose as trusted sources.



Cloud-Specific Security Challenges

- Multi-Tenancy Risks: Shared infrastructure increases risks like co-residence attacks.
- Data Location & Segregation: Data stored in shared environments across jurisdictions.
- Gartner's Seven Cloud Risks:
 - Privileged user access.
 - Regulatory compliance.
 - Data segregation.
 - Recovery challenges.
 - Investigative support limitations.
 - Long-term viability concerns.



Links : <https://aws.amazon.com/solutions/guidance/multi-tenant-architectures-on-aws/>

What are some real-world examples of denial-of-service (DoS) attacks?

- Dyn DDoS Attack (2016):

- A massive Distributed Denial-of-Service (DDoS) attack using the Mirai botnet targeted Dyn, a major DNS provider.
- Impact: Disrupted major websites like Netflix, Twitter, Reddit, and PayPal across the U.S. and Europe.
- Link : https://en.wikipedia.org/wiki/DDoS_attacks_on_Dyn

- GitHub Attack (2018):

- A record-breaking DDoS attack peaked at 1.35 Tbps using memcached servers for amplification.
- Impact: Temporarily knocked GitHub offline but was mitigated within minute.
- Link : <https://github.blog/news-insights/company-news/ddos-incident-report/>

- Cloudflare Attack (2020):

- Exploited weaknesses in the Network Time Protocol (NTP) for amplification, generating massive traffic loads.
- Impact: Demonstrated how protocol vulnerabilities could be leveraged for large-scale DDoS attacks.
- Link : <https://www.cloudflare.com/learning/ddos/famous-ddos-attacks/>

Security Mechanisms and Best Practices

- Policies: Define what is allowed or prohibited in a system. They act as the foundation of security by specifying acceptable behaviors, access controls, and operational guidelines.
- Mechanisms: Enforce the policies through tools and technologies, such as firewalls, encryption, and intrusion detection systems (IDS).

2. Network Security Steps

To secure cloud environments, organizations should follow these steps:

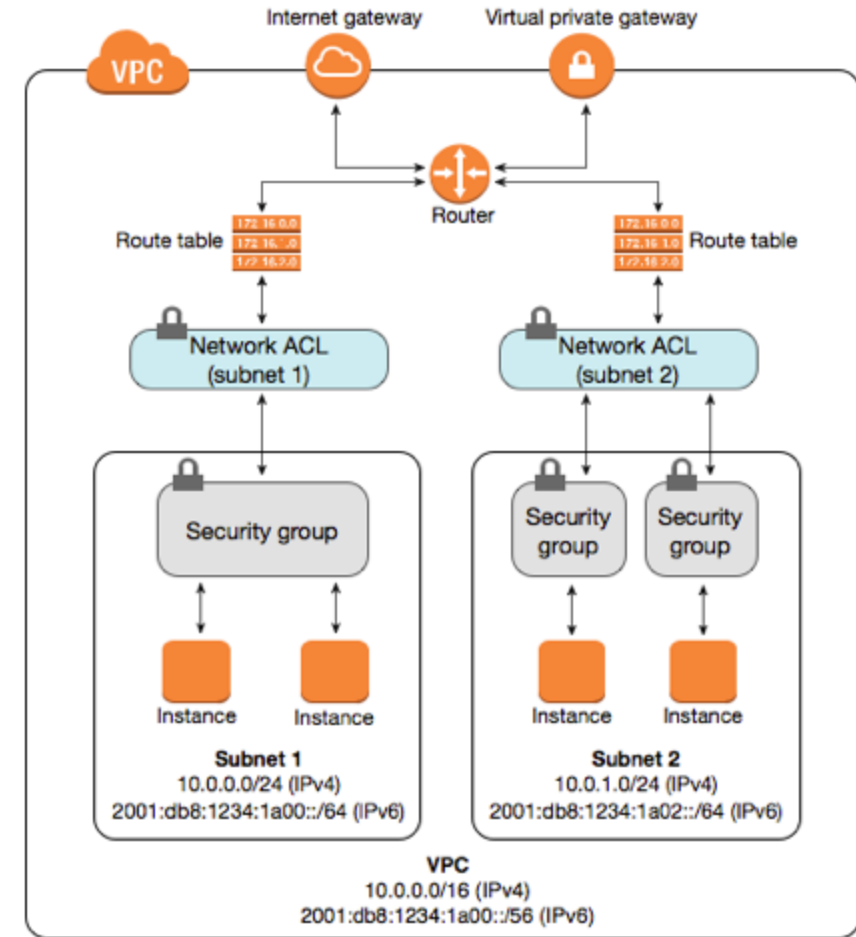
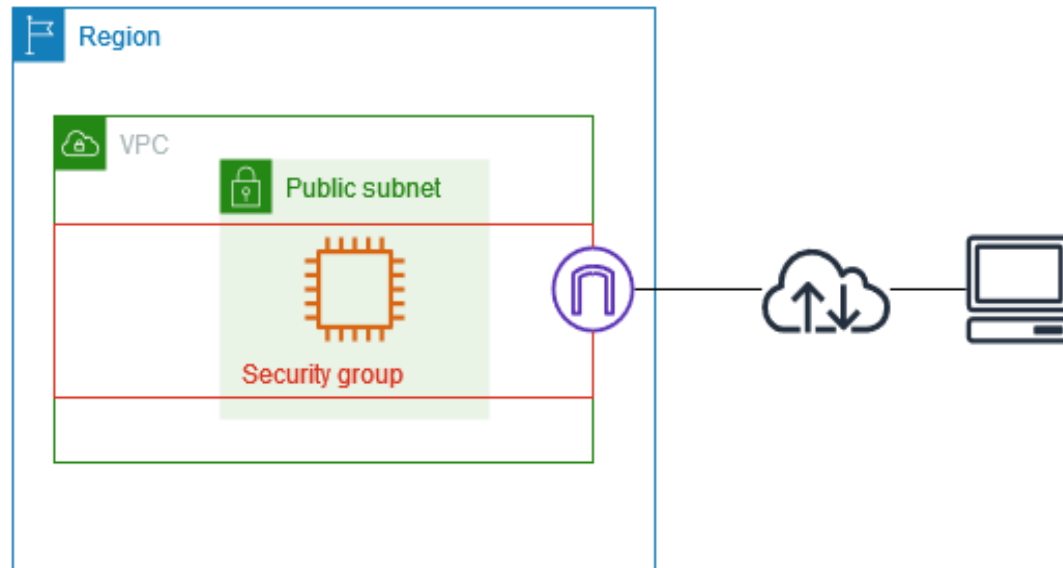
- Define Security Policy:
 - Establish a comprehensive policy covering usage, access controls, and data protection.
 - Include user training on password management and social engineering risks.
- Implement Firewalls and IDS:
 - Firewalls filter traffic based on predefined rules (e.g., IP addresses, ports).
 - IDS monitor network activity for suspicious patterns or breaches.
- Conduct Vulnerability Scanning and Penetration Testing:
 - Regularly scan for potential vulnerabilities using tools like Nessus or Metasploit.
 - Perform penetration testing to simulate attacks and identify weaknesses.

3. Virtualization Risks

Virtualization introduces unique risks due to the use of hypervisors (software managing virtual machines). Key vulnerabilities include:

- Hypervisor Exploits: Attackers can exploit bugs in the hypervisor to gain control of all hosted virtual machines (VMs).
- VM Escape: Malicious code can escape a VM's isolation layer to access the host system or other VMs.
- Rogue Hypervisors: Unauthorized hypervisors can be installed to control VMs covertly.
- Denial-of-Service (DoS): Overloading a hypervisor can disrupt all hosted VMs.

Security Measures in AWS



Link : <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-security-groups.html>

Hands-on Session Checklist

Step 1

- AWS Free-tier Account

Step 2

- IAM User
- MFA

Step 3

- Budget – zero spend

Step 4

- Hands-on EC2 – DONE
- S3 -

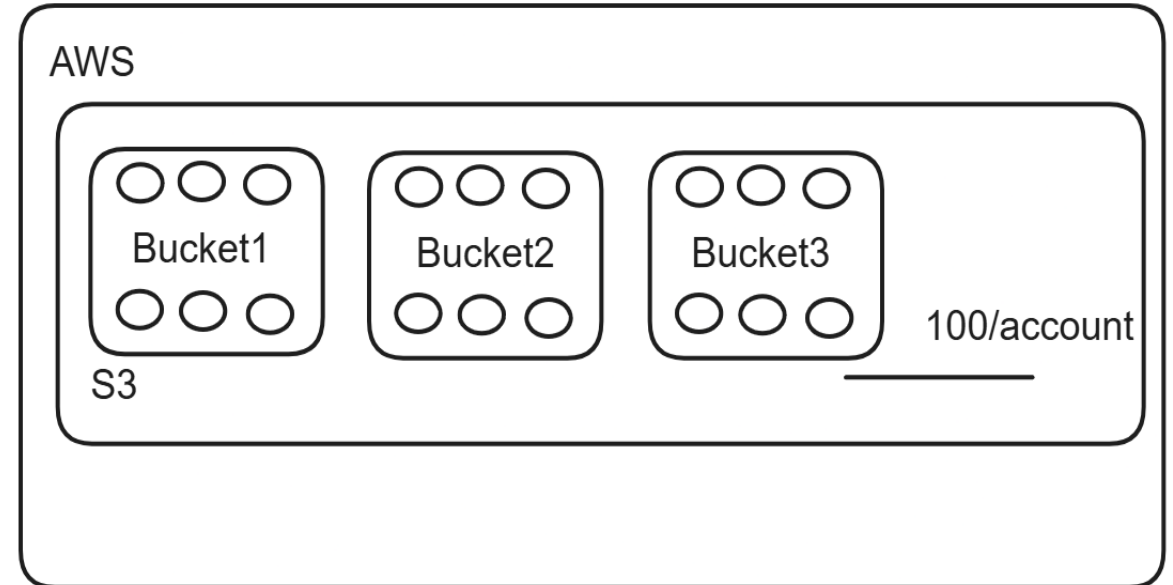
Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Use Free-tier
functions only

Features of Amazon S3

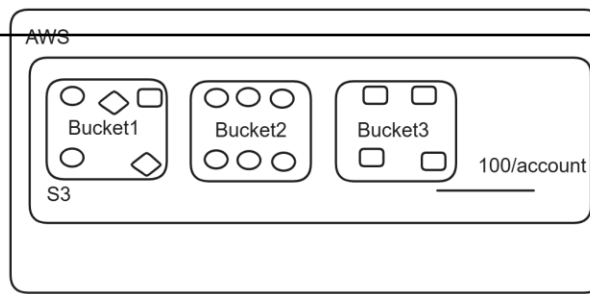
- **Bucket** : A general-purpose bucket is a container for objects stored in Amazon S3. You can store any number of objects in a bucket and all accounts have a default bucket quota of 10,000 general purpose buckets. Every object is contained in a bucket. For example, if the object named `photos/puppy.jpg` is stored in the `amzn-s3-demo-bucket` bucket in the US West (Oregon) Region, then it is addressable by using the URL <https://amzn-s3-demo-bucket.s3.us-west-2.amazonaws.com/photos/puppy.jpg>.
- When you create a bucket, you enter a bucket name and choose the AWS Region where the bucket will reside. After you create a bucket, you cannot change the name of the bucket or its Region. Bucket names must follow the [bucket naming rules](#). You can also configure a bucket to use [S3 Versioning](#) or other [storage management](#) features.
- **Object** : Objects are the fundamental entities stored in Amazon S3. Objects consist of object data and metadata. The metadata is a set of name-value pairs that describe the object. These pairs include some default metadata, such as the date last modified, and standard HTTP metadata, such as Content-Type. You can also specify custom metadata at the time that the object is stored.



<https://aws.amazon.com/ec2/pricing/on-demand/>

Buckets

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.



Use Free-tier
functions only

Features of Buckets

- When you create a general purpose bucket, you choose its name and the AWS Region to create it in. After you create a general purpose bucket, you can't change its name or Region. The following sections provide information about general purpose bucket naming, including naming rules, best practices, and an example for creating a general purpose bucket with a name that includes a globally unique identifier (GUID).
- Important Rules :
 - Bucket names **must** be unique across all AWS accounts in all the AWS Regions within a partition. A partition is a grouping of Regions. AWS currently has three partitions: aws (commercial Regions), aws-cn (China Regions), and aws-us-gov (AWS GovCloud (US) Regions).
 - A bucket name can't be used by another AWS account in the same partition until the bucket is deleted. **After you delete a bucket, be aware that another AWS account in the same partition can use the same bucket name for a new bucket and can therefore potentially receive requests intended for the deleted bucket.** If you want to prevent this, or if you want to continue to use the same bucket name, don't delete the bucket. We recommend that you empty the bucket and keep it, and instead, block any bucket requests as needed.

• Bucket policy

• First delete objects and then delete Bucket

• <https://aws.amazon.com/s3/pricing/>

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Principal": {
        "AWS": "arn:aws:iam::111122223333:role/JohnDoe"
      },
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:GetObjectVersion"
      ],
      "Resource": "arn:aws:s3:::amzn-s3-demo-bucket/*",
      "Condition": {
        "StringEquals": {
          "s3:ExistingObjectTag/environment": "production"
        }
      }
    }
  ]
}
```

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html#BasicsBucket>

Hands-on Session

Let's Create a S3 Bucket Policy

<https://awspolicygen.s3.amazonaws.com/policygen.html>